

Topic 6: Estimation

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Statistics (III) Midterm

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Core Questions

- What makes an estimator a “good” estimator?
- When population parameters are unknown, how do we estimate them from a sample?
- Why does the quality of an estimator directly affect SPC monitoring performance?

Outlines:

- 1 Theoretical Definitions
- 2 Mathematical Derivation
- 3 Statistical Interpretation
- 4 SPC Applications

Why Is Estimation Important in SPC?

In SPC, the in-control state of a process is usually described by baseline parameters such as μ_0, σ_0

However, in practice these quantities are usually **unknown** and must be estimated from Phase I historical data

Key points:

- If $\hat{\mu}_0$ is estimated incorrectly, the center line will shift
- If $\hat{\sigma}_0$ is estimated incorrectly, the control limits will become too wide or too narrow
- Incorrect limits may lead to:
 - Too many false alarms
 - Delayed detection of abnormal changes

Therefore, SPC is not only about drawing charts. The first question is:

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How good is the estimator?

Estimators and Sampling Distributions

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} F_\theta$$

where θ is an unknown parameter

An **estimator** is a function of the sample:

$$\hat{\theta} = T(X_1, \dots, X_n)$$

Since $X_1, \dots, X_n$ are random, $\hat{\theta}$ is also a random variable

Therefore, we should not only ask “what value do we get?” , but also ask:

- What is $E(\hat{\theta})$?
- What is $\text{Var}(\hat{\theta})$?
- Does $\hat{\theta}$ converge to the true value θ as $n \rightarrow \infty$?

Bias: Does the Estimator Systematically Deviate from the Truth?

Definition:

$$\text{Bias}(\hat{\theta}, \theta) = E(\hat{\theta}) - \theta$$

If for all θ

$$E(\hat{\theta}) = \theta$$

then $\hat{\theta}$ is called an **unbiased estimator**

Interpretation: Bias measures whether the estimator tends to overestimate or underestimate the true value on average

Example: Suppose $E(X_i) = \mu$ and $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Then $E(\bar{X}) = \mu$

Therefore, $\bar{X}$ is an unbiased estimator of μ

MSE: Unbiasedness Is Not Enough

The **Mean Squared Error** is defined as

$$\text{MSE}(\hat{\theta}, \theta) = E[(\hat{\theta} - \theta)^2]$$

It can be **decomposed** into

$$\text{MSE}(\hat{\theta}, \theta) = \text{Var}(\hat{\theta}) + \text{Bias}^2(\hat{\theta}, \theta)$$

Key message:

- An estimator can be unbiased but still have very large variance
- A slightly biased estimator may have a much smaller MSE

Example: Sample Proportion

目標: 估計 μ , 假設 $X_1, \dots, X_n \sim (\mu, \sigma^2)$

考慮兩個估計量: $\hat{\mu}_1 = \bar{X}$, $\hat{\mu}_2 = 0.8\bar{X}$

第一個估計量: $E(\hat{\mu}_1) = \mu$, $\text{Var}(\hat{\mu}_1) = \frac{\sigma^2}{n}$, 則 $\text{MSE}(\hat{\mu}_1) = \frac{\sigma^2}{n}$

第二個估計量:

$$E(\hat{\mu}_2) = 0.8\mu, \quad \text{Var}(\hat{\mu}_2) = 0.8^2 \text{Var}(\bar{X}) = 0.64 \frac{\sigma^2}{n}$$

$$\text{且 } \text{Bias}(\hat{\mu}_2, \mu) = 0.8\mu - \mu = -0.2\mu,$$

$$\text{則 } \text{MSE}(\hat{\mu}_2) = 0.64 \frac{\sigma^2}{n} + (0.2\mu)^2$$

結論: biased estimator 仍可能因 variance 較小而有更低的 MSE

Consistency: 樣本變大時, 估計會不會靠近真值?

定義: 若估計量序列 $\hat{\theta}_n$ 滿足

$$\hat{\theta}_n \xrightarrow{P} \theta$$

則稱 $\hat{\theta}_n$ 是 θ 的 **consistent estimator**

也就是對任意 $\varepsilon > 0$: $P(|\hat{\theta}_n - \theta| > \varepsilon) \rightarrow 0 \quad (n \rightarrow \infty)$

直觀意義:

- 小樣本時, 估計量可能誤差很大
- 當樣本越來越多, 估計量應該越來越穩定地貼近真值

一個常見的充分條件: $E(\hat{\theta}_n) \rightarrow \theta$ 且 $\text{Var}(\hat{\theta}_n) \rightarrow 0$,

例子: Bernoulli

設

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(\pi)$$

其中

$$P(X_i = 1) = \pi, \quad P(X_i = 0) = 1 - \pi$$

樣本比例 $\hat{p} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

性質:

$$E(\hat{p}) = \pi$$

所以 $\hat{p}$ 是 unbiased estimator。

又因為

$$\text{Var}(X_i) = \pi(1 - \pi),$$

故

$$\text{Var}(\hat{p}) = \frac{\pi(1 - \pi)}{n} \rightarrow 0$$

從估計到 SPC: Phase I 的核心

在 SPC 的 Phase I, 我們需要先建立 in-control baseline

以 $\bar{X}$ -chart 為例, 若已知 μ_0, σ_0 , 控制界限可寫成

$$UCL = \mu_0 + z_{1-\alpha/2} \frac{\sigma_0}{\sqrt{m}}, \quad LCL = \mu_0 - z_{1-\alpha/2} \frac{\sigma_0}{\sqrt{m}}$$

但實務上通常未知, 因此改用估計量: $\hat{\mu}_0, \hat{\sigma}_0$

則實際的控制界限為

$$\widehat{UCL} = \hat{\mu}_0 + z_{1-\alpha/2} \frac{\hat{\sigma}_0}{\sqrt{m}}, \quad \widehat{LCL} = \hat{\mu}_0 - z_{1-\alpha/2} \frac{\hat{\sigma}_0}{\sqrt{m}}$$

考慮問題:

- 若 $\hat{\mu}_0$ 有偏, 則中心線位置偏移
- 若 $\hat{\sigma}_0$ 變異大, 則界限不穩定
- 若估計不一致, 則無法建立可靠基準

估計量品質如何影響監控：以 p-chart 為例

若每批樣本大小為 n , 不良率為 π , 則

$$\hat{p} = \frac{X}{n}, \quad X \sim \text{Binomial}(n, \pi)$$

在 Phase I 中, 通常用 $\bar{p}$ 作為 π_0 的估計
對應的 p-chart 控制界限常寫為

$$UCL = \bar{p} + z\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}, \quad LCL = \bar{p} - z\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

Conclusion

(1) A good estimator should not be judged only by unbiasedness

- $\text{MSE} = \text{Var} + \text{Bias}^2$

(2) Consistency is a requirement for long-run reliability

- $\hat{\theta}_n \xrightarrow{P} \theta$

(3) In SPC, estimation is the foundation of monitoring

- Phase I requires estimation of μ_0 and σ_0 in order to determine control limits
- Control limits determine false alarm rates and detection power

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If the baseline estimation is unreliable,
then the entire SPC monitoring system becomes unreliable.

謝謝聆聽

備用頁: 可能 Q&A

Q1. 為什麼 unbiased 不一定最好?

因為 unbiased 只表示平均不偏, 但若 variance 很大, 估計仍可能很不穩。因此更完整的衡量是

$$\text{MSE} = \text{Var} + \text{Bias}^2$$

Q2. consistency 和 unbiased 有什麼差別?

- unbiased: 有限樣本下的平均正確性
- consistency: 大樣本下是否收斂到真值

Q3. 這和 SPC 最直接的連結是什麼?

Phase I 必須估計 in-control baseline ; 估計品質會直接影響 control limits, 進而影響監控表現。